

Customer segmentation with clustering

Name : Ishwarya Thanigaivelan

Date : 27/01/2025

Table of Content

- 1.Introduction
- 2.Problem Statement
- 3.Data description(EDA)
- 4.Methods and Results
 - 4a. Elbow method
 - 4b. Silhouette
 - 4c. K means Clustering
- 5. Boxplot for the clusters
- 6. Dimensional Reduction and Visualisation using PCA and t-SNE
- 6.Conclusion

Introduction

Segmentation of customers was done using various methods in order to group the customer for better marketing. Therefore, Data exploration, feature engineering, dimension reduction, and customer segmentation with clustering models were performed.

Problem Statement

Developing a robust customer segmentation using various clustering methods to assist the e-commerce company in understanding and serving its customers better. This will help to have a more customer-centric focus, improving marketing efficiency and increasing the revenue by targeting the customer groups.

Data description(EDA)

Data provided has 951,668 rows and 20 columns. Data was preprocessed, missing and duplicate values were identified. New features were created. Scaling was done using a standardisation method. EDA was performed. The features were grouped into one customer per row to aggregate the rows. Features are standardized using standard scaler. After scaling was performed to detect anomaly using one class SVM method. Found to be 5%. The size of the sample was reduced to prevent crashing.

Methods

To predict the value of cluster(k), the Elbow method and Silhouette score were performed. Dendrogram was created by performing Hierarchical clustering. The number of clusters predicted by all the three methods was 5.

Elbow method

The Elbow method was tried with different cluster values. The graph is shown in Figure 1.

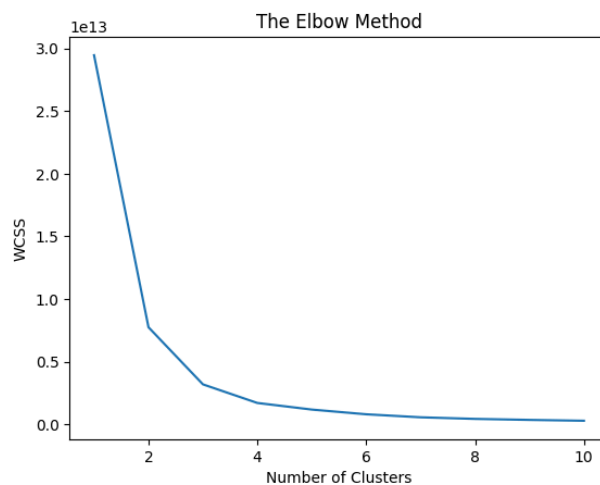


Figure 1: Elbow method for Number of cluster

Results

There is a bend in the graph between 4 and 6. It shows 5 would be an optimum for clustering the customers.

Silhouette score Method

The Silhouette score was calculated to find the number of clusters, which is best for grouping the customers. Silhouette coefficient values and Scatter plot was shown in Figure 2.

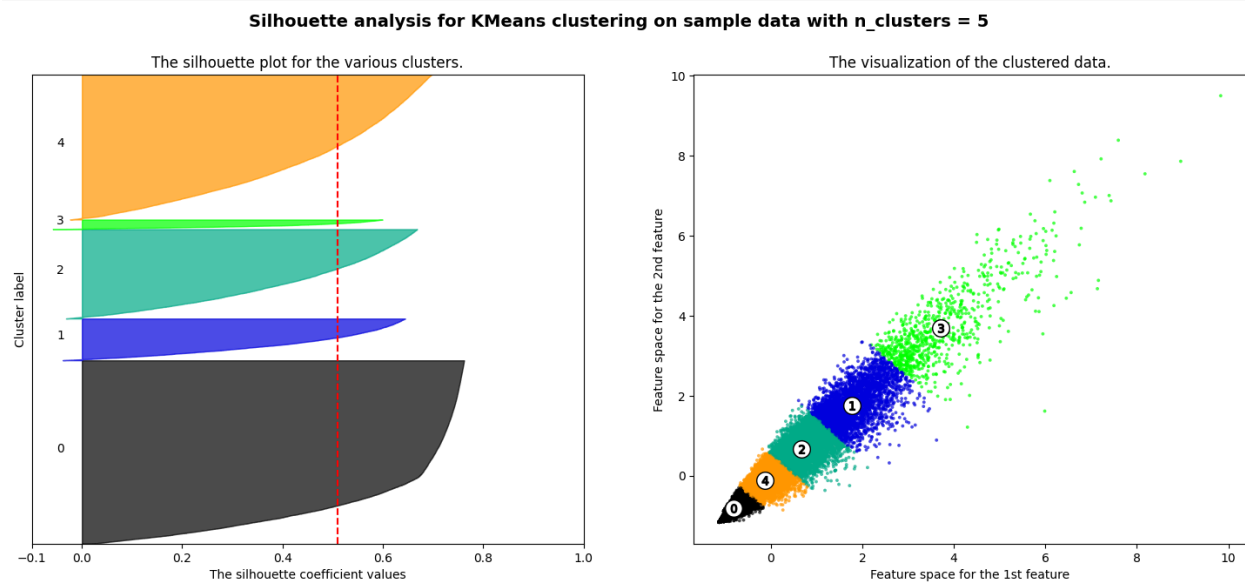


Figure 2: Silhouette Analysis on sample data with n_clusters = 5.

Results

The value of two clusters in silhouette is method 0.6223660646936714. There is no big variation in silhouette scores of clusters. The value shows data points are clustered better. The best number of clusters would be 5.

Hierarchical clustering Method

Dendrogram was created. It shows, 5 would be the optimum number for cluster.

Results

Dendrogram was shown in Figure 3.

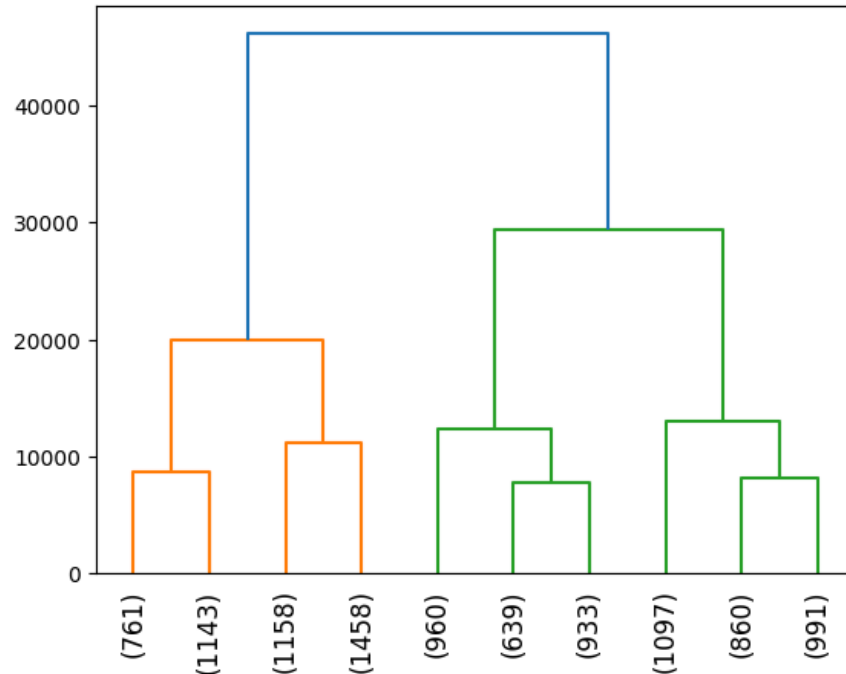


Figure 3: Dendrogram

K means clustering

K means clustering was performed by taking 5 as the optimum number of clusters from all the above methods. Scatter plots were created by keeping Total revenue in x- axis and all the other features are plotted against it. Scatter plot for Total revenue against frequency is shown in Figure 4 and Plot for Total revenue against recency is shown in Figure 5.

Result

In some of the Scatter plots the clusters are overlapped and some have clear clusters.

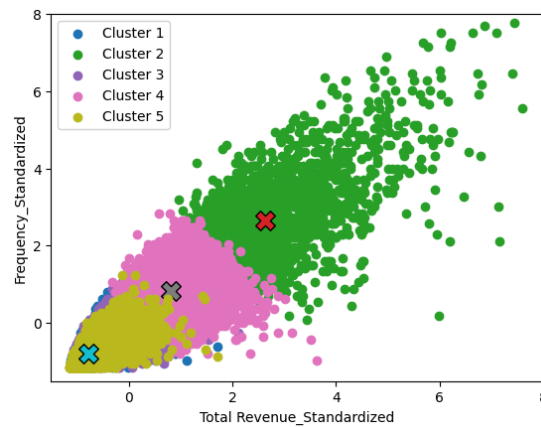


Figure 4: Scatter plot for Frequency

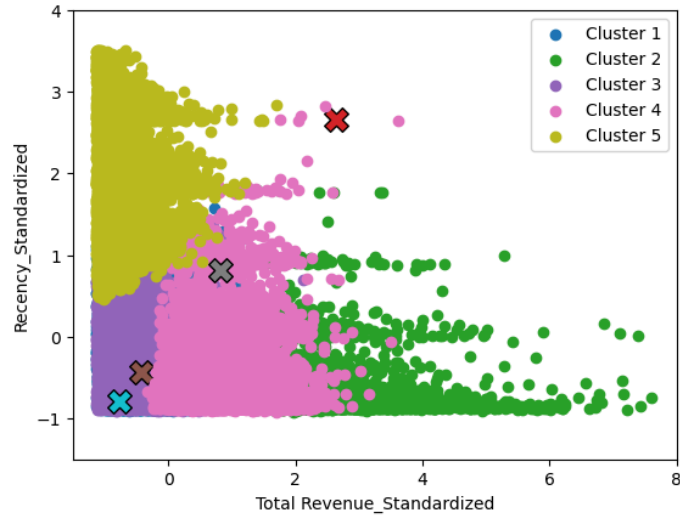


Figure 5: Scatter plot for Recency

Box Plots for Clusters and Features

Data frame for Cluster Association with each customer ID. Box plots are created for clusters and customer ID with regard to the features (Total revenue, Average unit cost, Frequency, Recency and age). Box plots for Total revenue is shown in figure 6, Box plots for Average unit cost is shown in figure 7, Box plots for frequency is shown in figure 8, Box plots for Recency is shown in figure 9 and Box plots for Age is shown in figure 10 respectively.

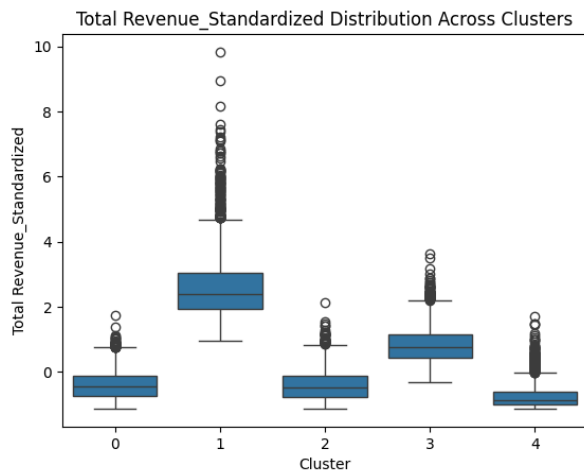


Figure 6: Boxplot for Total Revenue

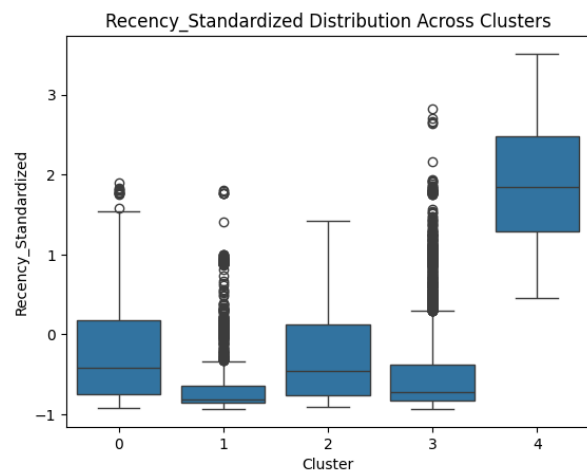


Figure 9: Boxplot for Recency

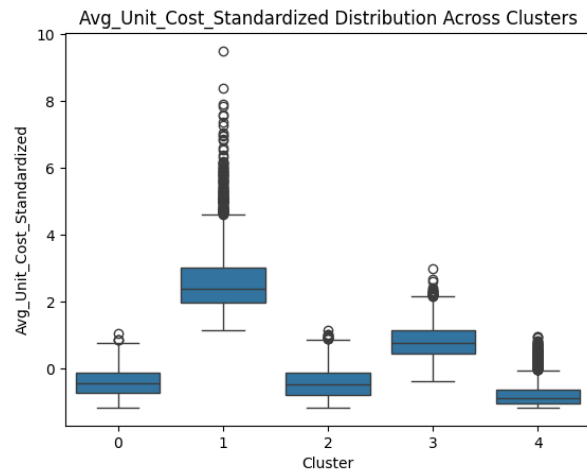


Figure 7: Box plot for Average unit cost

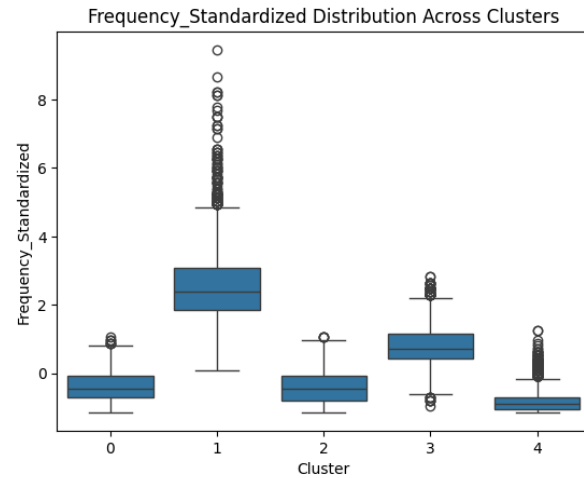


Figure 8: Box plot for Frequency

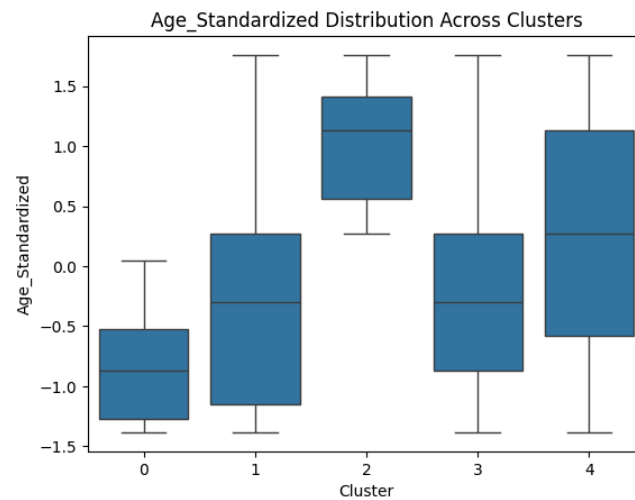


Figure 10: Box plot for Age

Results and Discussion

The box plot for total revenue, average unit cost and frequency looks more less similar. When looking into recency and age there is a difference in the size of the box. In total revenue, average unit cost and frequency, cluster 1 shows broader box than other clusters thus indicating large variability in cluster within the respective features.

The Box plot for recency show difference in median line is box, which shows, it could be an important discriminating feature for those clusters.

The box plot for age shows a wide box indicating a large spread in the data, meaning the feature has high variability within that cluster.

5. Dimensional Reduction and Visualisation using PCA and t-SNE

Dimensionality reduction and Visualisation using PCA

Five features were reduced to two components using PCA and merged with clusters to create a scatter plot to analyze visually. 2D visualisation for PCA is shown in Figure 11.

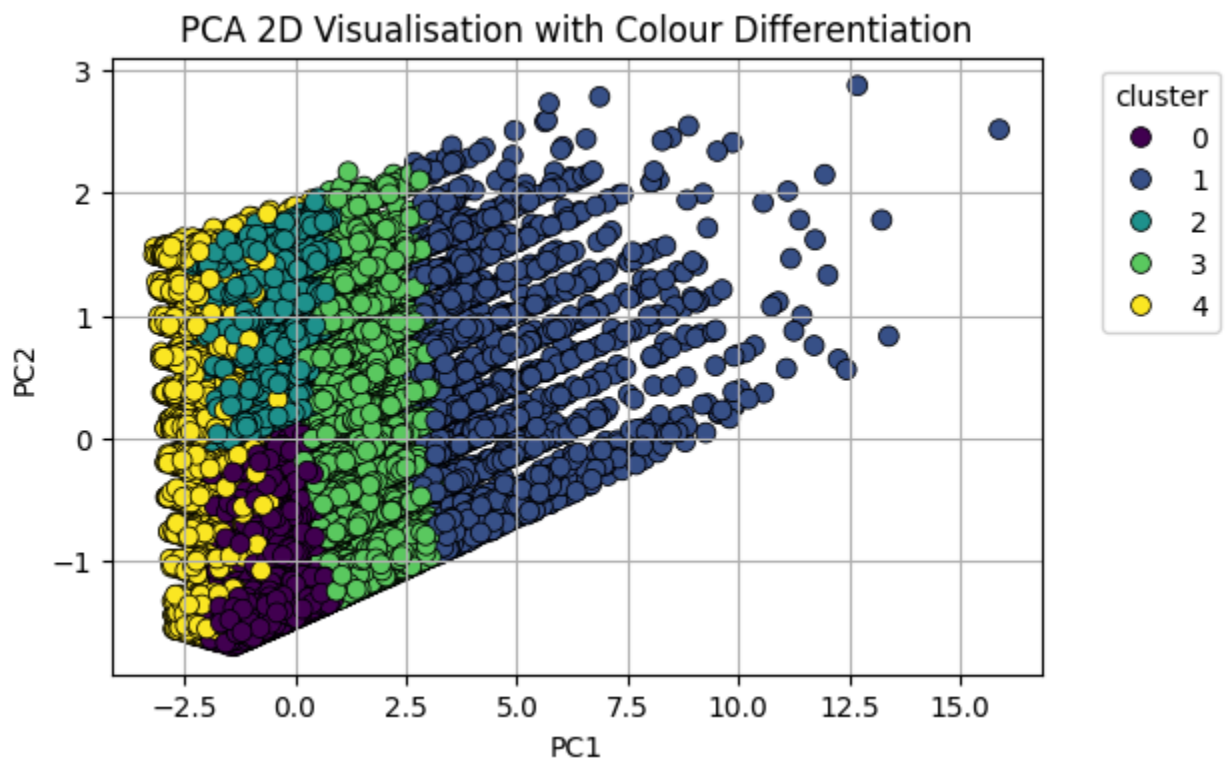


Figure 11: 2D Visualisation for PCA

Results

The 2D visualization of PCA shows the clusters are merged with one another.

Dimensionality reduction and Visualisation using t-SNE

The principal component 1 and 2 were created by reducing dimension using t-SNE. The 2D scatter plot was created and shown in figure 12.

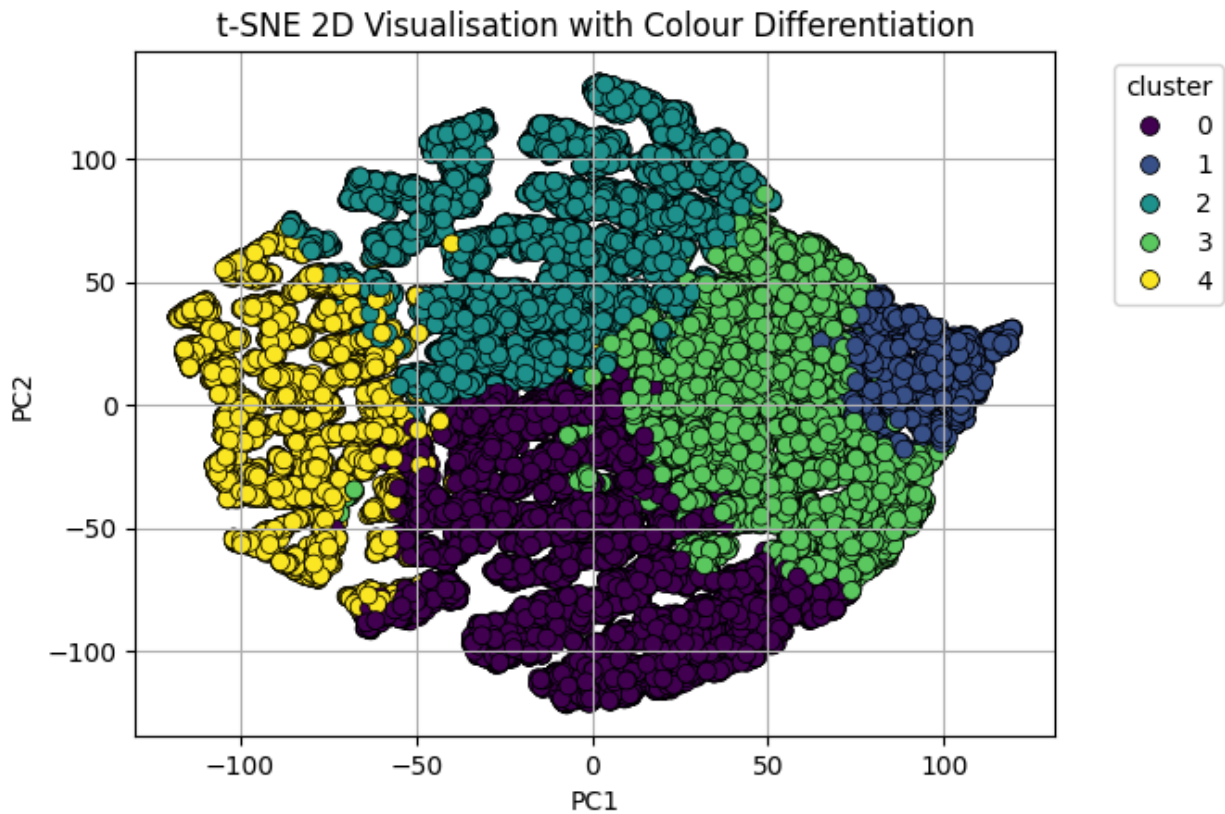


Figure 12: 2D Visualisation for t-SNE

Results

The 2D scatter plot shows better clustering than one done with PCA. Thus t-SNE is better for dimension reduction and 2D viewing of clusters.

Conclusion

By performing various clustering methods, it is found to be 5 is the best number for clustering the customer. Based on this, customer clustering can be done. Cluster 1 is well grouped and easily targeted for marketing.